

Benjamin (Ben) Zhou

YMSC, Tsinghua University, Jingzhai, 201

Email: byzhou01@gmail.com

Website: <https://byzhou01.github.io/>

EDUCATION/EMPLOYMENT

YMSC, Tsinghua University

2024-2027

Postdoc

Advisor: Shing-Tung Yau

Department of Mathematics, Northwestern University

9/2018 - 2024

Ph.D. in Mathematics

M.S. in Mathematics, en route

2019

Advisor: Eric Zaslow

Department of Computer Science, Stanford University

2015-2017

M.S. in Computer Science

Dual Concentration in Artificial Intelligence, Theoretical Computer Science

Advisor: Serafim Batzoglou

Department of Mathematics, Stanford University

2012-2016

B.S. in Mathematics, with Honors

Advisor: András Vasy

RESEARCH INTERESTS

My research interests are in Algebraic Geometry, Mathematical Physics, Symplectic Geometry, and artificial intelligence, quantum computing, mathematical finance, mathematical biology.

PUBLICATIONS

- *Enumerative geometry of quantum periods*, joint with Tim Gräfnitz, Helge Ruddat, and Eric Zaslow, Accepted to Advances in Mathematics, arXiv:2502.19408.
- *Tropical super Gromov-Witten invariants*, joint with Artan Sheshmani, and Shing-Tung Yau, submitted, arXiv:2510.17400.
- *Higher genus Gromov-Witten invariants from projective bundles on smooth log Calabi-Yau pairs*, submitted, arXiv:2503.17713.
- *Disentangling Direct from Indirect Relationships in Association Networks*, Xiao, N., A. Zhou, M. L. Kempher, B. Zhou, Z. Jason Shi, M. Yuan, X. Guo, L. Wu, D. Ning, Ml Kl Firestone, and J. -Z. Zhou, Proc. Nat. Acad. Sci. 119, No. 2, e2109995119.
- *Illumina sequencing-based analysis of free-living bacterial community dynamics during an Akashiwo sanguine bloom in Xiamen sea, China*, C Yang, Y Li, B Zhou, Y Zhou, W Zheng, Y Tian, JD Van Nostrand, L Wu, Scientific Reports 5 (1), 8476, 2015.
- *Functionally graded, bone-and tendon-like polyurethane for rotator cuff repair*, DFE Ker, D Wang, AW Behn, ETH Wang, X Zhang, BY Zhou, Advanced Functional Materials 28 (20), 1707107
- *Random sampling process leads to overestimation of beta-diversity of microbial communities*, J Zhou, YH Jiang, Y Deng, Z Shi, BY Zhou, K Xue, L Wu, Z He, Y Yang, MBio 4 (3), 10.1128/mbio. 00324-13

- *Metagenomic-based analysis of biofilm communities for electrohydrogenesis: From wastewater to hydrogen*, C Varrone, JD Van Nostrand, W Liu, B Zhou, Z Wang, F Liu, Z He, L Wu, International Journal of Hydrogen Energy 39 (9), 4222-4233

HONORS AND AWARDS

Huiyan Talent Fund, YMSC	2024-2027
Grade of "Excellent" in postdoc evaluation, YMSC	2024-2025
"Excellence as Graduate Teaching Assistant" Departmental Award	2021-2022
NSF RTG: "Analysis on Manifolds" recipient, National Science Foundation	2019
Semifinalist, Siemens Competition in Math, Science, and Technology	2012
National Finalist, Siemens Competition in Math, Science, and Technology	2011

TEACHING

Qiuzhen Yiyu Scholar, Tsinghua University	Spring 2026
Qiuzhen Yiyu Scholar, Tsinghua University	Spring 2025
TA for Math 351: Fourier Analysis and Boundary Value Problems, Northwestern University	Fall 2023
TA for Math 240: Linear Algebra, Northwestern University	Fall 2023
TA for Math 368: Introduction to Optimization, Northwestern University	Spring 2022
TA for Math 340: Geometry, Northwestern University	Spring 2022
TA for Math 366-1: Mathematical Models in Finance, Northwestern University	Winter 2022
TA for Math 336-1: Introduction to the Theory of Numbers, Northwestern University	Winter 2022
TA for Math 342: Introduction to Differential Geometry, Northwestern University	Fall 2021
TA for Math 320-1: Real Analysis, Northwestern University	Fall 2021
TA for Math 300: Foundations of Higher Mathematics, Northwestern University	Spring 2021
TA for Math 230-2: Multivariable Integral Calculus, Northwestern University	Winter 2021
TA for Math 230-1: Multivariable Differential Calculus, Northwestern University	Fall 2020
TA for Math 230-1: Multivariable Differential Calculus, Northwestern University	Fall 2019
TA for CS 229: Machine Learning, Stanford University	Fall 2016

INVITED TALKS

Fifty Years of Geometric Analysis and the Celebration of the Proof of the Calabi Conjecture, CUHK	
<i>Tropical super Gromov-Witten invariants</i>	4/15/26
Symplectic Geometry and Mathematical Physics seminar, BICMR	
<i>Higher genus open/closed Gromov-Witten correspondences for log Calabi-Yau surfaces</i>	11/11/25
Derived and Noncommutative Geometry seminar, SIMIS	
<i>Higher genus open/closed Gromov-Witten correspondences for log Calabi-Yau surfaces</i>	5/20/25
Algebraic Geometry seminar, BIMSA	
<i>Higher genus open/closed Gromov-Witten correspondences for log Calabi-Yau surfaces</i>	2/13/25
Symplectic Geometry seminar, YMSC	
<i>Higher genus open/closed Gromov-Witten correspondences for log Calabi-Yau surfaces</i>	10/16/24
Algebraic Geometry Seminar, University of Georgia	
<i>Higher genus open/closed Gromov-Witten correspondences for log Calabi-Yau surfaces</i>	11/1/2023
Geometry/Physics Seminar, Northwestern University	
<i>Higher genus open/closed Gromov-Witten correspondences for log Calabi-Yau surfaces</i>	10/26/2023
Graduate Student Seminar, Northwestern University	
<i>Curve Counting and Modular Forms</i>	6/2023

Graduate Student Seminar, Northwestern University

Every smooth cubic surface has exactly 27 lines

5/2021

CONFERENCES ATTENDED

ICCM, Shanghai	1/2-1/8/2026
String-Math, BIMS	2025
Topological Moonshine at UIUC	7/17-7/21/23
Los Angeles Workshop on Representations and Geometry at USC	6/12-6/16/23
Richmond Geometry Meeting at VCU	6/2-6/4/23
IHÉS Summer School in Enumerative Geometry, Physics, Representation Theory	7/2021

SERVICE AND OUTREACH

Tutoring of probability, linear algebra, real analysis to Northwestern undergraduates	2020-2023
---	-----------

PREVIOUS WORK EXPERIENCE

Research Assistant, Stanford Undergraduate Research in Computer Science	2016
Research Assistant, Stanford Undergraduate Research in Mathematics	2015
Software Engineering Intern, xAd Inc.	2015

MISCELLANEOUS

Programming languages: Python, C++, Java, Sage
Languages: English (native), Chinese (working proficiency)